# Lecture 21 (153 - 248)

## ARIMA Modeling

### ARMA(p, q)

$$AR(p): \quad y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

$\varepsilon_t$ shock

$$\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$$

ARMA(p, q)

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \eta_t$$

$$\eta_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

$$\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$$

(p=0) ARMA(0, q) $\quad y_t = \underbrace{\phi_0 + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots \theta_q \varepsilon_{t-q}}_{MA(q)}$

(q=0) AR(p)

$$y_t - \mu = \phi_1 (y_{t-1} - \mu) + \phi_2 (y_{t-2} - \mu) + \cdots + \phi_p (y_{t-p} - \mu) + \eta_t$$

$$\mu (1 - \phi_1 - \cdots - \phi_p) = \phi_0$$

$$\mu = \frac{\phi_0}{1 - \phi_1 - \cdots - \phi_p}$$

ARIMA

# BACKSHIFT NOTATION

$$B y_t = y_{t-1}, \quad B^2 y_t = y_{t-2}, \quad B^3 y_t = y_{t-3}$$

$$B^0 = I, \quad I y_t = y_t$$

$$\phi(z) = 1 + 2z - 3z^2$$

$$\phi(B) = I + 2B - 3B^2$$

$$\phi(B)\, y_t = y_t + 2y_{t-1} - 3y_{t-2}$$

$$(y_t - \mu) = \phi_1 (y_{t-1} - \mu) + \phi_2 (y_{t-2} - \mu) + \cdots + \phi_p (y_{t-p} - \mu)$$
$$+ \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$$

$$\theta(z) = 1 + \theta_1 z + \theta_2 z^2 + \cdots + \theta_q z^q$$

$$\phi(B)(y_t - \mu) = \theta(B)\varepsilon_t \longrightarrow \text{ARMA}(p, q) \text{ model}$$

Differencing Operator: $\nabla = I - B$

$$\nabla y_t = (I - B) y_t = y_t - y_{t-1}$$

## ARIMA Model

$$\nabla y_t = y_t - y_{t-1}$$

$$\nabla^2 y_t = \nabla (y_t - y_{t-1})$$

**Defn:** $y_t$ is ARIMA$(p, d, q)$ if $\nabla^d y_t$ is ARMA$(p, q)$

$$\phi(B)(\nabla^d y_t - \mu) = \theta(B)\varepsilon_t$$

$\rightarrow$ ARIMA$(p, d, q)$

$d = 1, 2$

$= (y_t - y_{t-1}) - (y_{t-1} - y_{t-2})$

$= y_t - 2y_{t-1} + y_{t-2}$

$d = 1$

$y_t - y_{t-1}$ is ARMA$(p, q)$

$y_t$

**Estimation**

① Write likelihood $\rightarrow$ $f_{y_1, \ldots y_n}$ $\rightarrow$ KALMAN FILTER

② Optimize likelihood

ARIMA

ARIMA$((p, 1, q), \underbrace{\text{trend} = 't'}_{d})$

$\mu = 0$

$d = 1$

$$\phi(B)(y_t - y_{t-1} - \mu) = \theta(B)\varepsilon_t$$

$$y_t = \cancel{\text{linear function of } t} + \text{some stochastic model term}$$

# AIC & BIC

$$AIC = -2 \times \text{maximized log-likelihood} + 2 \times \#\text{parameters}$$

$$BIC = -2 \times \text{maximized log-likelihood} + (\log n_{eff}) \times \#\text{parameters}$$

# Seasonal ARMA models

ARIMA( data, order = (p, d, q) ...)

SARIMAX Results

## AR(1):

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t$$

or $$(y_t - \mu) = \phi_1 (y_{t-1} - \mu) + \varepsilon_t$$

$$y_t = \phi_0 + \phi_1 y_{t-12} + \varepsilon_t$$ → Seasonal AR(1) with period 12

ARMA(1,1)

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

Seasonal ARMA(1,1)

$$y_t = \phi_0 + \phi_1 y_{t-12} + \varepsilon_t + \theta_1 \varepsilon_{t-12}$$

## ARMA (p, q)

$$\phi(B)\,(y_t - \mu) = \theta(B)\,\varepsilon_t$$

## Seasonal ARMA (P, Q) model

$$\Phi(B^{12})\,(y_t - \mu) = \Theta(B^{12})\,\varepsilon_t$$

(s) $$\Phi(B^s)(y_t - \mu) = \Theta(B^s)\,\varepsilon_t$$

→ Seasonal ARMA (P, Q)

## Multiplicative Seasonal ARMA

$$(y_t - \mu)$$

Seasonal AR(1) → $$y_t - \mu = \phi_1 (y_{t-12} - \mu) + \eta_t$$

$$\eta_t = \underline{\phi}_1 \eta_{t-1} + \varepsilon_t$$

$$\eta_t = y_t - \mu - \phi_1 (y_{t-12} - \mu)$$

$$= \phi(B^{12})\,(y_t - \mu)$$

$$\phi(B^{12})\,(y_t - \mu) = \underline{\phi}_1\, \phi(B^{12})(y_{t-1} - \mu) + \varepsilon_t$$

$$\underline{\phi}(B)\,\phi(B^{12})(y_t - \mu) = \varepsilon_t$$

ARMA $(p,q)$ model : $\phi(B)(y_t - \mu) = \theta(B)\varepsilon_t$

Seasonal ARMA $(P,Q)_s$ model : $\Phi(B^s)(y_t - \mu) = \Theta(B^s)\varepsilon_t$

Multiply:

$$\Phi(B^s)\phi(B)(y_t - \mu) = \Theta(B^s)\theta(B)\varepsilon_t$$

→ Multiplicative Seasonal ARMA model

## SARIMA

$y_t$

$$\Phi(B^s)\phi(B)(\nabla_s^D \nabla^d y_t - \mu) = \Theta(B^s)\theta(B)\varepsilon_t$$

↑ SARIMA model

ARIMA$(p,d,q) \times (P,D,Q)_s$

$D = 1$, $d = 1$